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Physics-Constrained Optimization Framework for Detecting Stealthy Drift Perturbations

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Abstract

This work develops a zero-trust, physics-constrained mathematical framework for detecting stealthy drift perturbations in power system dynamical models. Such perturbations constitute adversarial, statistical deviations that preserve first-order operating trends, making them difficult to identify using classical residual-based estimators or unconstrained data-driven models. We introduce ZETWIN, a spatio-temporal learning architecture formulated as a constrained optimization problem in which the nodal admittance matrix Y_{bus} acts as a graph-structured linear operator embedded directly into the loss functional. This construction enforces Kirchhoff-consistent latent representations and yields a mathematically grounded zero-trust decision rule that flags any trajectory violating physical feasibility, independent of prior attack signatures. The proposed framework is evaluated using a PyPSA-based AC–DC meshed network, demonstrating an AUROC = 0.994, and F1 = 0.969. The formulation highlights how physics-informed constraints, graph operators, and spatio-temporal approximation theory can be combined to construct mathematically interpretable zero-trust detectors for complex dynamical systems.

Keywords: detection; dynamical models; optimization; spatio-temporal; zero-trust

MSC: 97Rxx

1. Introduction

Modern electric power systems operate as an integrated Cyber–Physical Systems (CPS), where real-time telemetry, automated control, and distributed energy resources (DERs) interact across heterogeneous communication networks [1,2]. This increasing digitalization enhances efficiency and situational awareness, but also enlarges the attack surface. Among the most challenging threats are stealthy zero-day attacks [3–5], adversarial perturbations engineered to mimic legitimate operating conditions. Such attacks can remain hidden from both classical State Estimation (SE) residual checks and contemporary data-driven Intrusion Detection Systems (IDS), enabling adversaries to manipulate system behavior while evading detection.

Existing detection mechanisms fall broadly into two categories, each with fundamental limitations in a CPS context. Signature-based IDS, which relies on known attack patterns [6–8] and therefore fails against novel or adaptive adversaries. Statistical anomaly detectors, including many machine learning (ML) approaches [9–14], identify deviations from historical data distributions but are vulnerable to false alarms during legitimate grid fluctuations such as DER variability, topology changes, or N-1 contingencies. More

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critically, purely data-driven models treat grid telemetry as unstructured numerical sequences [15–17], ignoring the physical laws, such as Kirchhoff's Current Law (KCL) and Ohm's Law, that govern feasible power flows. This disconnect allows an attacker to inject measurements that appear statistically normal yet violate the underlying physics, thereby bypassing both SE and ML-based detectors.

To address this physics gap, we introduce ZETWIN (Zero-Trust Digital Twin), a zero-trust, physics-informed anomaly detection framework that shifts the focus from statistical irregularity to physical infeasibility. The central idea is to embed the network's admittance operator directly into the learning architecture, enabling the model to internalize the structural constraints of the grid. By incorporating a normalized admittance matrix into the loss functional, ZETWIN forms a physics-informed neural network (PINN) that penalizes latent representations that are inconsistent with power flow invariants. This operator-based constraint provides a mathematically grounded mechanism for rejecting trajectories that deviate from the feasible manifold defined by the grid topology.

The contributions of this work are summarized as follows:

- A spatio-temporal convolutional neural network (CNN) is developed to capture both temporal evolution and inter-bus electrical dependencies. The model integrates a normalized admittance operator into its latent space, enforcing Kirchhoff-consistent structure during training.
- An anomaly detection is formulated as a constrained optimization problem in which a trajectory is flagged as anomalous if it violates either statistical consistency or physical feasibility. This yields a principled decision rule combining probabilistic inference with operator-norm constraints.
- A loss functional is defined to show that the physics term induces contraction of latent embeddings toward the nullspace of Y_{norm}, corresponding to physically feasible states.
- A zero-day stealthy drift attack is modeled as smooth, low-amplitude perturbations, and its detectability is analyzed under the proposed zero-trust criterion.
- Using a PyPSA (Python for Power System Analysis) framework [18] of a meshed AC–DC network, we demonstrate that ZETWIN achieves a discriminative performance compared to classical SE and modern deep learning baselines.

Together, these contributions establish ZETWIN as a mathematically principled and operationally robust framework for anomaly detection in cyber-physical power networks. By grounding detection in the immutable laws of power system physics, ZETWIN provides a foundation for securing next-generation, inverter-dominated grids against sophisticated and stealthy adversaries.

The remainder of this paper is organized as follows. Section 2 presents the related works, Section 3 presents the mathematical formulation of the ZETWIN framework, including the construction of the digital twin, the derivation of the normalized admittance operator, and the physics-constrained learning architecture. Section 4 details the computational procedures used to instantiate these operators in practice, covering data generation, attack injection, and model training. Section 5 describes the simulation environment and benchmarking protocol, while Section 6 analyzes the detection dynamics and compares ZETWIN against baseline methods. Finally, Section 7 summarizes the key findings and outlines directions for future research in physics-informed, zero-trust anomaly detection for power system telemetry.

2. Related Work

This section situates the proposed ZETWIN framework within the broader landscape of physics-informed machine learning, neural operator theory, state estimation, digital twins, and cyber-physical security. While these research areas provide essential foundations,

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none directly address the problem of zero-trust, physics-constrained anomaly detection for stealthy drift attacks in power networks.

2.1. Physics-Informed Machine Learning and Neural Operators

Physics-informed machine learning (PIML) has emerged as a principled approach for embedding physical laws into data-driven models. The work in [19] provides a comprehensive overview of PIML techniques that incorporate differential equations, conservation laws, and operator constraints into neural architectures, demonstrating improved robustness and interpretability in noisy or data-sparse regimes. Similarly, ref. [20] presents case studies in weather and climate modeling where Partial Differential Equation (PDE)-based constraints stabilize high-dimensional spatio-temporal learning. These works highlight the value of physics-guided regularization but focus primarily on forward modeling and forecasting rather than adversarial anomaly detection.

A parallel line of research develops neural operators as infinite-dimensional analogues of neural networks. The work in [21] introduces the Multipole Graph Neural Operator for parametric PDEs, capturing nonlocal interactions through graph-based representations. The Neural Operator/Graph Kernel Network framework of [22] further demonstrates how operator learning can approximate solution maps across parameter spaces. While these methods showcase the power of graph-structured operators for modeling physical systems, they are designed for surrogate modeling and do not address the detection of malicious deviations from physically feasible trajectories.

Recent physics-informed and physics-constrained learning approaches have demonstrated the value of embedding physical knowledge into deep models for power and energy applications. Examples include physics-informed voltage stability monitoring [23], deep learning with physical model constraints and attention mechanisms for state prediction [24], physically consistent day-ahead dispatch and comfort control [25], and physics-constrained photovoltaic forecasting enhanced with signal decomposition [26]. Additional work has explored physics-constrained robustness evaluation for intelligent security assessment [27], ensemble-learning-based safety risk monitoring in cyber-physical power systems [28], and uncertainty-aware deep learning for health monitoring in safety-critical energy systems [29]. While these studies incorporate physical structure into learning models, they focus on prediction, forecasting, stability assessment, or safety monitoring. None formulates anomaly detection as a zero-trust, operator-constrained optimization problem. In contrast, ZETWIN embeds the normalized admittance operator directly into the latent space and uses it to define a feasibility-based decision rule that flags any trajectory violating Kirchhoff-consistent structure. This operator-level integration, combined with spatio-temporal learning, distinguishes ZETWIN from prior physics-informed or hybrid models (data-driven approaches) and specifically targets stealthy drift attacks that preserve first-order statistical trends.

A substantial body of work has focused on improving the robustness, accuracy, and cyber-resilience of power system state estimation (SE). The work in [30] proposed a Phasor Measurement Unit (PMU)-based robust SE method that enhances real-time monitoring by mitigating the impact of measurement noise and outliers, demonstrating the value of high-frequency PMU data for improving observability and estimation accuracy. The work in [31] investigates parameter error identification in SE, showing how modeling inaccuracies and parameter uncertainties can degrade estimation performance. While these contributions strengthen SE reliability under nominal or noisy conditions, they do not explicitly address adversarial scenarios in which an attacker injects measurements that remain physically plausible yet malicious.

To counter adversarial manipulation, ref. [32] introduces a replay-attack-resilient SE scheme that detects time-shifted measurement injections by exploiting temporal consistency. Although effective against replay-style attacks, this approach does not generalize to smooth, stealthy drift attacks engineered to remain statistically inconspicuous. Such attacks are increasingly relevant in modern cyber-physical systems, where adversaries exploit temporal structure to evade classical residual-based detectors.

Parallel to advances in SE, digital twin technologies have emerged as high-fidelity platforms for simulation, monitoring, and control. The work in [33] develops an ontological modeling framework for constructing power system digital twins, emphasizing semantic consistency and modularity. The work in [34] surveys digital twins for microgrids, highlighting their potential for real-time monitoring, predictive maintenance, and resilience analysis. While these works underscore the importance of digital twins as experimental testbeds, they do not integrate physics-informed learning or adversarial detection mechanisms into the twin itself. In contrast, ZETWIN leverages the PyPSA framework not only for simulation but also as a structural component of the detection pipeline, embedding the admittance operator directly into the learning process.

2.2. Anomaly Detection, Zero-Trust Security, and Adversarial Robustness

From the machine learning perspective, ref. [35] provides a comprehensive survey of deep learning methods for time-series anomaly detection, covering supervised, unsupervised, and hybrid approaches. Although these methods excel at identifying statistical irregularities, they typically lack explicit physical constraints and are therefore vulnerable to attacks that remain within the statistical envelope of normal behavior. This limitation motivates the need for anomaly detection frameworks that combine temporal learning with operator-based feasibility checks.

Cybersecurity research has increasingly emphasized zero-trust principles and adversarial robustness. In [36], they proposed a flexible zero-trust architecture for industrial IoT infrastructures, advocating continuous verification and minimal implicit trust. The work in [37] surveys adversarial machine learning attacks and defenses, demonstrating that deep learning models are susceptible to carefully crafted perturbations. The work in [38] introduces constrained adversarial machine learning (ConAML), a framework for CPS that incorporates physical constraints into both attack and defense formulations. These works highlight the importance of integrating physical structure into cybersecurity mechanisms; however, they do not provide a unified anomaly detection framework that couples spatio-temporal learning with power network operators.

Finally, ref. [39] constrained overcomplete analysis operator learning for cosparse signal modeling, showing how operator constraints can shape latent representations. Their insights inform the design of ZETWIN's physics-constrained latent space, where the normalized admittance matrix acts as a feasibility operator enforcing Kirchhoff-consistent embeddings.

These gaps motivate the ZETWIN framework, which integrates digital twin simulation, spatio-temporal learning, and operator-based physical constraints into a unified zero-trust anomaly detection methodology. In summary, while prior work has established the value of physics-informed learning, neural operators, and CPS security analysis, there remains a gap at their intersection: a mathematically grounded, operator-constrained, zero-trust anomaly detection framework tailored to stealthy drift attacks in power networks. ZETWIN is designed to occupy this niche by embedding the admittance operator into the learning process, coupling data-driven anomaly scores with physics residuals, and demonstrating their joint effectiveness in a digital twin environment.

Notation

Table 1 presents the symbols and meaning used throughout the paper.

Table 1. Summary of notation used throughout the ZETWIN framework.

Symbol	Meaning
$\mathcal{B}$	Set of buses in the power network
$\mathcal{L}$	Set of Alternating Current (AC) and Direct Current (DC) transmission lines
N	Number of buses
z_{ij}	Complex impedance of line (i,j)
y_{ij}	Line admittance $1/z_{ij}$
Y	Nodal admittance matrix
Y_{norm}	Normalized admittance matrix
$x(t)$	True system state at time t
$\tilde{x}(t)$	Noisy PMU measurement
X_t	Sliding telemetry window
$a(t)$	Stealthy drift attack signal
$\mathcal{D}_{\text{clean}}$	Clean dataset for training
Φ_θ	Spatio-temporal CNN
$z_\theta(X_t)$	Latent representation
$p_\theta(X_t)$	Anomaly probability
$\mathcal{L}_{\text{phys}}$	Physics-constrained loss
$\mathcal{L}_{\text{BCE}}$	Binary cross-entropy loss
λ	Physics-constraint weight
τ, δ	Zero-trust decision thresholds

3. ZETWIN Pipeline: Mathematical Formulation and Zero-Trust Workflow

This section presents a mathematical formulation of the ZETWIN pipeline in Figure 1. Each stage transforms raw power system telemetry into structured representations suitable for zero-trust anomaly detection. The workflow integrates graph-theoretic operators, constrained optimization, and spatio-temporal learning, ensuring that the detection mechanism remains grounded in the physical laws governing AC–DC power networks.

3.1. Digital Twin Construction and Admittance Matrix Derivation

Let $\mathcal{B} = \{1, \dots, N\}$ denote the set of buses and $\mathcal{L} = \mathcal{L}_{\text{AC}} \cup \mathcal{L}_{\text{DC}}$ the set of AC and DC transmission elements. The digital twin is represented as a graph

$$\mathcal{G} = (\mathcal{B}, \mathcal{L}), \qquad \mathcal{L} \subseteq \mathcal{B} \times \mathcal{B}, \tag{1}$$

where each line $(i,j) \in \mathcal{L}$ has complex impedance

$$z_{ij} = r_{ij} + jx_{ij}, \qquad y_{ij} = \frac{1}{z_{ij}}. \tag{2}$$

The nodal admittance matrix $Y \in \mathbb{C}^{N \times N}$ is defined by

$$Y_{ij} = \begin{cases} \sum\limits_{k \in \mathcal{N}(i)} y_{ik}, & i = j, \\ -y_{ij}, & (i,j) \in \mathcal{L}, \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

To improve numerical conditioning during training, we normalize its magnitude:

$$Y_{\text{norm}} = \frac{|Y|}{\max_{i,j} |Y_{ij}|}. \tag{4}$$

Since $|Y|$ denotes the elementwise magnitude of the complex admittance matrix, Y_{norm} is a real-valued operator in $\mathbb{R}^{N \times N}$, preserving the graph structure while removing phase information.

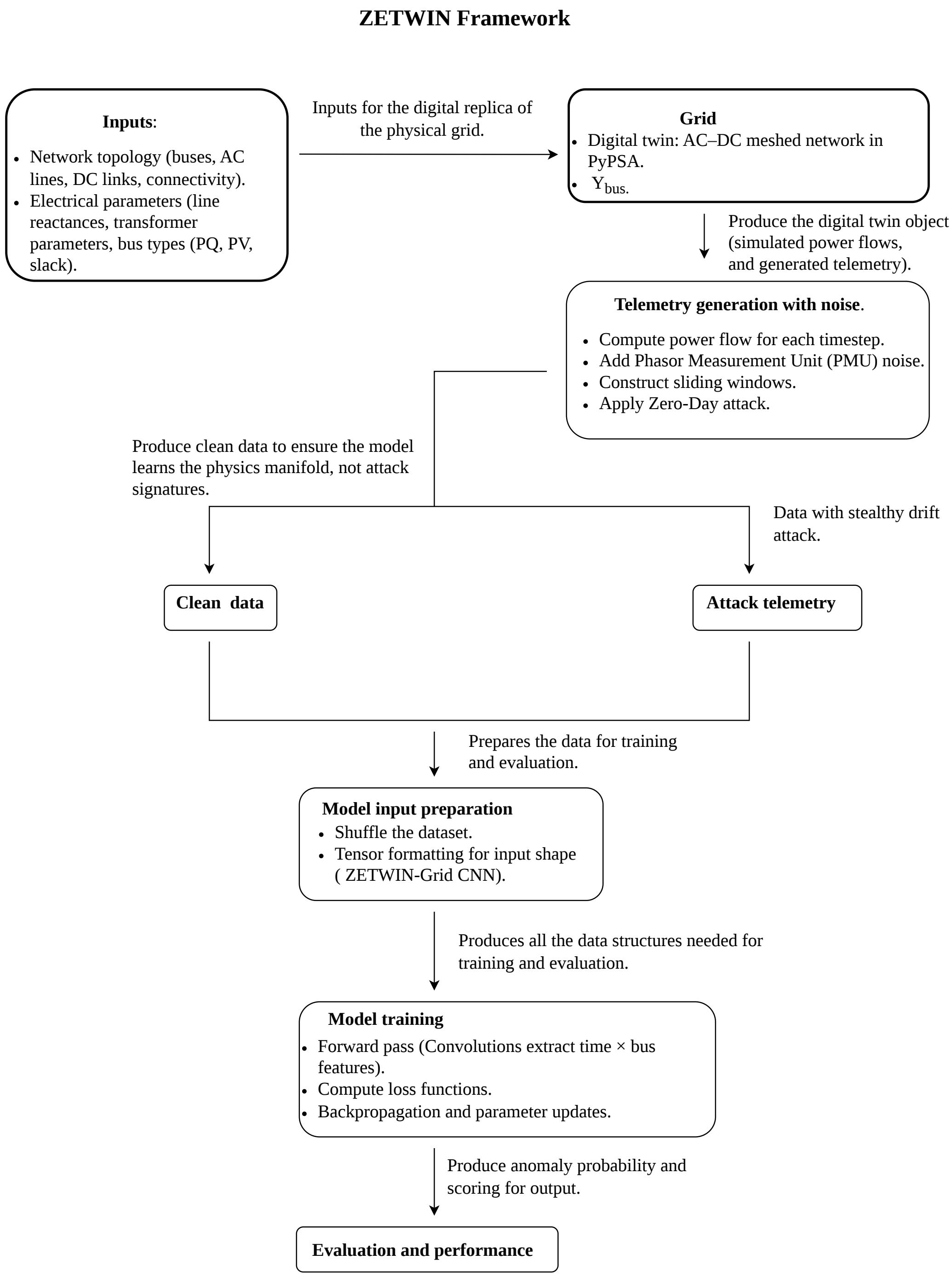

Figure 1. End-to-end data-generation pipeline for ZETWIN. The digital twin produces clean system trajectories, PMU noise is added, stealthy drift attacks are injected, and sliding windows are constructed for training and evaluation. This figure describes the simulation and preprocessing workflow.

The PyPSA framework is used to create the digital twin $\mathcal{D}$, which instantiates the network graph in (1) with line parameters from (2) and uses (3) to compute AC–DC power flows, generating physically consistent system states that emulate the behavior of the real grid.

3.2. Telemetry Generation and Noise Modeling

At each time index t, the digital twin solves the nonlinear AC–DC power flow equations

$$f_{\text{PF}}\big(V(t), \theta(t), P(t), Q(t)\big) = 0, \tag{5}$$

yielding the system state

$$x(t) = \big[V(t), \theta(t), P(t), Q(t)\big], \tag{6}$$

where $V(t)$ and $\theta(t)$ denote the bus voltage magnitudes and phase angles, and $P(t)$ and $Q(t)$ denote the active and reactive power injections, respectively.

PMU measurements are modeled as noisy observations:

$$\tilde{x}(t) = x(t) + \eta(t), \qquad \eta(t) \sim \mathcal{N}(0, \sigma^2 I), \tag{7}$$

where $\eta(t)$ is a random noise vector added to the true system state $x(t)$, and σ is selected to match typical PMU noise levels (0.01–0.05 p.u.) for voltage magnitude measurements.

To capture temporal structure, we construct sliding windows of length T:

$$X_t = \big[\tilde{x}(t - T + 1), \ldots, \tilde{x}(t)\big] \in \mathbb{R}^{T \times N_f}. \tag{8}$$

3.3. Analytical Zero-Day Stealthy Drift Attack Model

A zero-day stealthy drift attack is modeled as a smooth, low-amplitude perturbation

$$a(t) = \epsilon\, g(t), \tag{9}$$

where $g(t)$ is a slowly varying function and ϵ controls the drift magnitude.

The attacked telemetry is

$$x_{\text{atk}}(t) = x(t) + a(t), \tag{10}$$

subject to the stealthiness condition

$$\|a(t)\|_2 \ll \|x(t)\|_2, \tag{11}$$

ensuring that the attack remains statistically inconspicuous.

3.4. Zero-Trust Data Preparation

Under the zero-trust principle, never trust, always verify, the learning process assumes that no telemetry can be trusted unless it is explicitly verified as clean. Accordingly, the model is trained exclusively on clean data:

$$\mathcal{D}_{\text{train}} = \mathcal{D}_{\text{clean}}. \tag{12}$$

After batching, the tensors take the form

$$X \in \mathbb{R}^{B \times T \times N_f}, \qquad y \in \{0, 1\}^B, \tag{13}$$

where B is the batch size, and y is the corresponding binary label vector.

3.5. Spatio-Temporal CNN for Graph-Structured Telemetry

In Figure 2, we present the ZETWIN model architecture, highlighting the spatio-temporal CNN, the physics-constrained latent space, and the dual zero-trust decision rule.

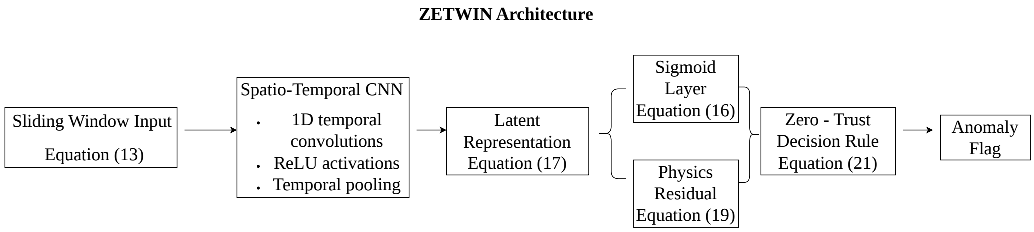

Figure 2. A spatio-temporal CNN processes a sliding PMU window X_t to produce a bus-aligned latent representation $z_\theta(X_t)$. The statistical path computes an anomaly probability $p_\theta(X_t)$, while the physics path evaluates the feasibility residual $r_t = \|Y_{\text{norm}} z_\theta(X_t)\|_2$. The dual zero-trust decision rule declares an anomaly when either the statistical detector or the physics constraint is violated.

Let Φ_θ denote the spatio-temporal CNN parameterized by θ, acting as a nonlinear operator

$$\Phi_\theta : \mathbb{R}^{T \times N_f} \to [0, 1]. \tag{14}$$

The first convolutional layer extracts temporal features:

$$h^{(1)} = \sigma\big(W^{(1)} * X_t + b^{(1)}\big), \tag{15}$$

where $*$ denotes convolution along the temporal dimension and σ is a Rectified Linear Unit (ReLU) activation. The final output is the anomaly probability

$$p_\theta(X_t) = \Phi_\theta(X_t). \tag{16}$$

The latent representation before the final layer is denoted

$$z_\theta(X_t) \in \mathbb{R}^d, \tag{17}$$

where $d = N$ is chosen to ensure dimensional compatibility with the admittance operator Y_{norm}.

3.6. Physics-Constrained Zero-Trust Loss Function

To enforce physical feasibility, we penalize latent embeddings that violate the admittance operator:

$$Y_{\text{norm}}\, z_\theta(X_t) \approx 0. \tag{18}$$

The physics loss is

$$\mathcal{L}_{\text{phys}}(\theta) = \|Y_{\text{norm}}\, z_\theta(X_t)\|_2^2. \tag{19}$$

The full zero-trust objective combines binary cross-entropy with the physics constraint:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{BCE}}(\theta) + \lambda\, \mathcal{L}_{\text{phys}}(\theta), \tag{20}$$

where $\lambda > 0$ controls the strength of the constraint.

Given that Y_{norm} is a graph-structured linear operator whose nullspace corresponds to Kirchhoff-consistent voltage patterns, the physics-constrained loss acts as a projection toward the feasible manifold. Minimizing (19) therefore induces a contraction of latent representations toward this manifold. Any stealthy drift attack that perturbs the system away from physical feasibility increases the operator residual, providing a theoretical guarantee that such deviations can not remain latent-space invariant. This establishes the operator residual as a physically grounded certificate of infeasibility and justifies its role in the zero-trust decision rule.

3.6.1. Admittance Constraint Improves Detection

The normalized admittance matrix Y_{norm} encodes the graph structure and Kirchhoff-consistent relationships of the underlying AC–DC network. Its nullspace $\ker(Y_{\text{norm}})$ corresponds to the set of voltage patterns that satisfy the network's current-balance equations, making (18) a necessary condition for physical feasibility. Any deviation from this nullspace reflects a violation of Kirchhoff's laws and therefore indicates that the trajectory cannot arise from a valid power-flow solution. By penalizing (19), the learning process contracts latent representations toward this feasibility manifold. Stealthy drift attacks, although statistically smooth, inevitably perturb the system in directions that move $z_\theta(X_t)$ away from $\ker(Y_{\text{norm}})$, causing the operator residual to increase. This yields a physically grounded anomaly signal that complements the learned spatio-temporal features, and improves detectability, compared with physics-informed approaches that incorporate physical knowledge indirectly, such as through PDE residual penalties, auxiliary physics-based features, or soft regularizers [40–43].

3.6.2. Dimensional Compatibility and Physical Meaningfulness

The admittance operator Y_{norm} acts on bus-level voltage patterns, and its nullspace characterizes the set of Kirchhoff-consistent states. For this reason, the latent representation is constructed to have N dimensions, ensuring that the product of (4) and (17) is physically meaningful; it corresponds to a current-balance residual across the network. This dimensional compatibility is essential, as it allows the physics loss to penalize deviations from the feasibility manifold $\ker(Y_{\text{norm}})$ rather than acting as an abstract regularizer. Stealthy drift attacks perturb the system in directions that move (17) away from this manifold, causing the operator residual to increase.

3.7. Zero-Trust Decision Rule

A trajectory is flagged as anomalous if either the learned probability exceeds a threshold τ or the physics violation exceeds a tolerance δ:

$$\text{Declare anomaly if} \quad p_\theta(X_t) > \tau \quad \text{or} \quad \|Y_{\text{norm}}\, z_\theta(X_t)\|_2 > \delta. \tag{21}$$

This dual criterion operationalizes the zero-trust principle: every window must independently verify both its statistical consistency and its physical feasibility, and any violation along either dimension triggers an anomaly declaration.

4. Methods

This section describes the computational procedures used to implement and evaluate the ZETWIN zero-trust detection framework. Whereas Section 3 formalized the mathematical operators underlying the pipeline, the present section details how these operators are instantiated in simulation, how telemetry is generated and processed, and how the learning and evaluation procedures are executed in practice.

4.1. Digital Twin Simulation and Power Flow Evaluation

The digital twin $\mathcal{D}$ is implemented using the PyPSA framework, which provides a physics-based simulation environment for AC–DC power networks. The network topology is instantiated from the graph defined in (1), with line parameters specified by (2). PyPSA constructs the full component model, including buses, AC lines, DC links, transformers, converters, and automatically assembles the corresponding nonlinear power-flow equations in (5). Using PyPSA's Newton–Raphson solver with default tolerances. This produces a physically consistent system state in (6), which serves as the ground-truth trajectory for telemetry generation.

To support the physics-constrained component of ZETWIN, the nodal admittance matrix Y is computed directly from the instantiated network using (3). Its normalized form Y_{norm} is obtained using (4), ensuring numerical stability during training. In the controlled simulation environment used for this study, the network topology remains fixed, and therefore Y and Y_{norm} are computed once at initialization. In real-world deployment, these operators would be updated dynamically by the Energy Management System (EMS) topology processor whenever a legitimate topology change occurs.

4.2. Telemetry Generation and Sliding Window Construction

To emulate realistic phasor measurement unit (PMU) telemetry, the clean system state $x(t)$ produced by the digital twin is corrupted with additive Gaussian noise in (7). The noise is applied independently to each feature of each bus, ensuring that the resulting telemetry reflects sensor-level uncertainty rather than structural perturbations.

Each measurement vector $\tilde{x}(t)$ contains N_f features per bus, including voltage magnitude, voltage angle, active power injection, and reactive power injection. These features are concatenated across all buses, yielding a time-indexed multivariate signal in $\mathbb{R}^{N_f}$.

To capture temporal dependencies, the noisy telemetry is converted into overlapping sliding windows of fixed length T using (8), where the window advances by one timestep. This procedure transforms the single-timestep PMU stream into a sequence of spatio-temporal tensors suitable for convolutional processing.

Each window X_t constitutes a single input sample to the ZETWIN model. During training, only windows derived from clean trajectories are used, consistent with the zero-trust protocol. During evaluation, windows may contain clean or attacked telemetry, enabling the model to detect deviations in both temporal evolution and instantaneous physical feasibility.

4.3. Stealthy Drift Attack Injection

To evaluate zero-day robustness, we inject a class of low-and-slow false-data injection attacks designed to mimic realistic sensor drift or gradual load bias. Each attack is constructed as a smooth, time-varying perturbation applied directly to the clean system state $x(t)$ generated in (9). In all experiments, ϵ is selected such that the resulting drift magnitude remains within 4–6% of nominal voltage levels, ensuring that the perturbation overlaps with natural PMU noise and load fluctuations.

The attacked state is then obtained by superimposing the drift onto the clean trajectory in (10). This formulation preserves the temporal smoothness of the underlying physical trajectory while gradually biasing the voltage magnitude of the targeted bus. The attack is initiated at a randomized start time $t_{\text{start}} \in [5, 15]$ to prevent the detector from overfitting to a fixed temporal pattern.

To ensure stealthiness, the perturbation is constrained to remain small relative to the nominal state in (11), which guarantees that the injected drift does not trigger classical residual-based detectors. This design reflects realistic zero-day adversarial behavior, where the attacker avoids abrupt deviations and instead introduces a gradual bias that accumulates over time [44,45]. The resulting attack telemetry is subsequently processed into sliding windows (Section 4.2) and passed to the ZETWIN model for evaluation.

4.4. Dataset Preparation and Zero-Trust Training Protocol

The full dataset consists of multivariate time-series trajectories generated by the digital twin, each representing either nominal operation or a zero-day stealthy drift attack. After telemetry generation (Section 4.2) and attack injection (Section 4.3), each trajectory is segmented into overlapping sliding windows, which serve as the fundamental input samples for learning and evaluation.

We partition the resulting dataset into two disjoint subsets:

$$\mathcal{D}_{\text{clean}}, \qquad \mathcal{D}_{\text{atk}}, \tag{22}$$

where $\mathcal{D}_{\text{clean}}$ contains only windows derived from nominal trajectories, and $\mathcal{D}_{\text{atk}}$ contains windows that include the slow drift perturbation. The two sets are kept strictly separate to preserve the zero-trust assumption.

Following the zero-trust training principle in (12), the ZETWIN model is trained exclusively on $\mathcal{D}_{\text{clean}}$. No attacked samples are used during training, hyperparameter tuning, or threshold selection. This ensures that the learned detector models only the distribution of physically feasible, attack-free telemetry and do not rely on prior knowledge of attack signatures.

Before training, the clean dataset is shuffled and grouped into minibatches of size B. Each batch takes the form in (13). During training, all labels in $\mathcal{D}_{\text{clean}}$ are set to $y = 0$, reflecting the absence of anomalies. Attacked samples and their labels ($y = 1$) are used only during evaluation to compute ROC, PR, F1, detection delay, and false-alarm metrics.

This dataset preparation pipeline ensures that the learning process is fully consistent with the zero-trust paradigm, the model is exposed only to clean telemetry during training, while all anomaly detection capabilities must emerge from the combination of spatio-temporal learning and the physics-constrained loss.

4.5. Spatio-Temporal CNN Architecture

The ZETWIN detector is implemented as a compact spatio-temporal convolutional neural network designed to extract both short-term temporal patterns and cross-bus correlations from PMU telemetry. The model defines a nonlinear mapping in (14).

The first convolutional layer applies one-dimensional temporal filters of width $k = 3$ across each feature channel in (15). This layer captures short-term temporal dependencies, such as local voltage trends and smooth drift patterns.

Subsequent convolutional and max-pooling layers progressively downsample the temporal dimension while increasing the number of feature channels, enabling the network to learn higher-level spatio-temporal abstractions. After the final convolutional block, the resulting tensor is flattened and passed through a fully-connected layer to produce a latent representation in (17). This design allows the latent vector to be interpreted as a bus-level voltage pattern in a learned feature space.

Finally, a sigmoid output layer maps the latent representation to an anomaly probability in (16).

4.6. Physics-Constrained Optimization

To ensure that the learned latent representation remains consistent with the underlying network physics, ZETWIN incorporates a physics-constrained loss based on the normalized admittance operator Y_{norm}. Since the nullspace of Y_{norm} corresponds to Kirchhoff-consistent voltage patterns, any physically feasible state must satisfy (18). The physics residual in (19) penalizes latent embeddings that deviate from the feasibility manifold $\ker(Y_{\text{norm}})$, effectively contracting the learned representation toward physically plausible states. Stealthy drift attacks, although statistically subtle, inevitably perturb the system in directions that increase this residual.

The weighted-sum formulation in (20) follows standard practice in physics-informed learning, where physical constraints are incorporated as soft penalties rather than as independent optimization objectives. In this setting, the binary cross-entropy and the physics residual are not competing objectives; $\mathcal{L}_{\mathrm{BCE}}(\theta)$ drives the classifier to separate clean and anomalous trajectories, while $\mathcal{L}_{\mathrm{phys}}(\theta)$ acts as a structural prior that contracts latent representations toward the feasibility manifold defined by Y_{norm}. Because the physics residual is computed using the normalized admittance operator, its scale remains stable across networks of different sizes, making the choice of λ transferable and preventing the loss from growing with grid size. Consequently, the model is not highly sensitive to small variations in λ, values within the same order of magnitude yield similar performance. The scalar weight, therefore, serves only to balance numerical scales for stable optimization.

4.7. Zero-Trust Evaluation and Decision Rule

During inference, ZETWIN evaluates each sliding window using both the learned anomaly probability and the physics residual. A window is flagged as anomalous if either the statistical detector or the physics-based feasibility check indicates a violation in (21).

This dual zero-trust criterion ensures robustness against zero-day attacks, the learned CNN captures subtle temporal deviations, while the physics residual provides a physically grounded certificate of infeasibility. As a result, ZETWIN detects both statistically stealthy drift attacks and physically inconsistent trajectories that would evade purely data-driven models.

4.8. Algorithmic Implementation

This section provides a complete algorithmic description of the ZETWIN zero-trust detection framework. Whereas Sections 3 and 4 introduced the mathematical operators, digital twin construction, telemetry generation, attack modeling, and learning procedures, the present section explains how these components are executed in practice. The workflow consists of four stages: digital twin initialization and admittance operator construction, dataset generation through power-flow simulation, noise injection and optional drift attacks, physics-constrained zero-trust training, and evaluation using the dual decision rule.

4.9. End-to-End ZETWIN Workflow

Algorithm 1 summarizes the full pipeline. The procedure begins by instantiating the PyPSA-based digital twin and constructing the admittance matrix Y using (3), followed by normalization via (4). Clean system trajectories are generated by repeatedly solving the AC–DC power flow equations (5). PMU noise is added according to (7) to obtain realistic telemetry. When generating attacked samples, a smooth drift signal $a(t)$ is injected using (9) and (10). Each trajectory is then segmented into sliding windows X_t using (8), which serve as inputs to the spatio-temporal CNN.

During training, the model computes anomaly probabilities using (16), latent embeddings using (17), and physics residuals using (18). The full loss (20) is minimized via stochastic gradient descent. During zero-day evaluation, the trained model produces both anomaly scores and physics residuals for each window, enabling the dual zero-trust decision rule in (21).

The ZETWIN workflow is designed so that every stage of the pipeline enforces the zero-trust principle and maintains consistency with the physical behavior of the underlying AC–DC network. The digital twin initialization ensures that all downstream computations are grounded in physically valid power-flow solutions, providing a reliable foundation for telemetry generation and operator construction.

During dataset generation, the framework produces both nominal and attacked trajectories. Clean trajectories reflect the true system dynamics obtained from the digital twin, while attacked trajectories incorporate smooth drift perturbations constructed to satisfy the stealthiness condition in (11). This separation preserves the zero-trust requirement that the model is trained exclusively on clean data.

Algorithm 1 ZETWIN: End-to-End Zero-Trust Detection Framework

Input: PyPSA network $\mathcal{G}$, window length T, drift magnitude ϵ, physics weight λ
Output: Trained model Φ_θ, anomaly scores, physics residuals
(A) Digital Twin Initialization
Load AC–DC PyPSA network $\mathcal{G}$
Construct admittance matrix Y using (3)
Normalize Y_{norm} using (4)
(B) Dataset Generation
for $n = 1$ to N_{samples} **do**
    Simulate clean state $x(t)$ via power flow (5)
    Add PMU noise using (7)
    if attack sample **then**
        Inject drift $a(t)$ using (9)
        Form attacked signal $x_{\text{atk}}(t)$ using (10)
    end if
    Construct sliding window X_t using (8)
end for
(C) Model Training
for each epoch **do**
    for each minibatch (X, y) **do**
        Compute prediction $p_\theta(X)$ using (16)
        Compute latent embedding $z_\theta(X)$ using (17)
        Compute physics residual $Y_{\text{norm}} z_\theta(X)$ using (18)
        Compute loss $\mathcal{L}(\theta)$ using (20)
        Update parameters $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}(\theta)$
    end for
end for
(D) Zero-Day Attack Simulation
Generate long-horizon trajectory with drift beginning at t_{start}
for $t = 1$ to $T_{\text{total}} - T$ **do**
    Extract window X_t
    Compute anomaly score $p_\theta(X_t)$
    Compute physics residual $\|Y_{\text{norm}} z_\theta(X_t)\|_2$
end for
Return: Trained model Φ_θ, anomaly scores, physics residuals

The sliding-window transformation converts raw PMU telemetry into fixed-length spatio-temporal tensors, enabling the CNN to learn temporal dependencies and cross-bus correlations. These windows form the fundamental input samples used throughout training and evaluation.

Physics-constrained training further reinforces feasibility by penalizing latent representations that violate the admittance operator, as defined in (18)–(20). This contraction toward the feasibility manifold ensures that the learned representation remains consistent with Kirchhoff's laws, even in the presence of statistically subtle perturbations.

Finally, zero-day evaluation applies the dual decision rule in (21), combining the learned anomaly probability with the physics residual. This guarantees that both statistical deviations and physically infeasible trajectories are detected, even when the attack remains smooth, gradual, and statistically inconspicuous.

Together, these components ensure that ZETWIN does not rely on prior attack signatures and remains robust to zero-day, low-and-slow manipulations that evade traditional data-driven or residual-based detectors.

ZETWIN is not trained as a supervised classifier. The model is trained exclusively on nominal (attack-free) PMU telemetry, and no attack labels are used at any stage of training or threshold selection. The CNN learns the structure of normal spatio-temporal voltage behavior, while the physics residual measures instantaneous deviations from the Kirchhoff consistent manifold. During inference, these two signals are combined to produce an anomaly score, and the detection threshold is chosen using a clean validation set. Because the framework relies solely on nominal data and does not require examples of attacks, it is fully compatible with the zero-day, zero-trust setting in which attack signatures are unavailable during training.

4.10. Computational Complexity and Deployment Considerations.

ZETWIN is designed to be computationally lightweight and suitable for real-time deployment in wide-area monitoring systems. The detection pipeline consists of a single forward pass through a compact spatio-temporal CNN (≈0.5 M parameters) followed by a physics residual computation using the sparse admittance operator. Both components scale linearly with the number of buses and PMU channels, and the admittance matrix operations exploit the inherent sparsity of transmission and distribution networks.

On a standard CPU, ZETWIN processes a full multivariate PMU window in under 1 ms, meeting real-time requirements for 30–120 Hz PMU data streams. Because the method operates on fixed-length temporal windows and relies on sparse linear algebra, it extends naturally to larger networks without requiring architectural changes. These properties make ZETWIN practical for integration into existing monitoring infrastructure, where it can operate alongside state estimation and bad-data detection modules.

4.11. Training Algorithms for Baseline Models

To benchmark ZETWIN against standard anomaly detection approaches, we train four learning-based baselines and implement one physics-based baseline. The learning models include: a CNN without physics constraints, an LSTM classifier, an Autoencoder, and the State-Estimation residual detector. All learning models follow the same minibatch training structure as Algorithm 1, with differences arising only in the loss function and model architecture.

4.11.1. Physics-Constrained ZETWIN-CNN

This model incorporates the physics loss (19) into the training objective. The latent representation is regularized to satisfy the admittance constraint, ensuring that embeddings remain close to the Kirchhoff-consistent feasibility manifold (see Algorithm 2).

Algorithm 2 Training ZETWIN(CNN with Physics-Constrained)

```
Input: Training set (X, y), normalized admittance Y_norm, physics weight λ
for each epoch do
    for each minibatch (X_b, y_b) do
        Compute prediction p_θ(X_b)
        Compute latent embedding z_θ(X_b)
        Compute physics residual r = Y_norm z_θ(X_b)
        Compute loss L = L_BCE + λ||r||_2^2
        Update θ ← θ − η∇_θ L
    end for
end for
```

4.11.2. CNN Without Physics Constraint

This baseline isolates the contribution of the physics term by training the same architecture using only the binary cross-entropy loss (see Algorithm 3).

Algorithm 3 Training CNN Without Physics Constraint

Input: Training set (X, y)
for each epoch **do**
  for each minibatch (X_b, y_b) **do**
    Compute prediction $p_\theta(X_b)$
    Compute loss $\mathcal{L} = \mathcal{L}_{\text{BCE}}$
    Update $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$
  end for
end for

4.11.3. LSTM Classifier

Each bus is treated as an independent univariate time series. This model captures temporal patterns but ignores spatial coupling and physical structure, providing a baseline for purely temporal anomaly detection (see Algorithm 4).

Algorithm 4 Training LSTM Classifier

Input: Training set (X, y), number of buses N
for each bus $b = 1, \ldots, N$ **do**
  Extract univariate series $X^{(b)}$
  for each epoch **do**
    for each minibatch $(X_b^{(b)}, y_b)$ **do**
      Compute prediction $p_\theta(X_b^{(b)})$
      Compute loss $\mathcal{L} = \mathcal{L}_{\text{BCE}}$
      Update $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$
    end for
  end for
end for

4.11.4. Autoencoder

This unsupervised baseline reconstructs each bus's time series and flags anomalies using reconstruction error,

$$\text{score}(X_b) = \|X_t - \hat{X}_t\|_2. \tag{23}$$

It provides a comparison point for reconstruction-based methods that do not use physics constraints or spatial coupling (see Algorithm 5).

Algorithm 5 Training Autoencoder

Input: Training set X, number of buses N
for each bus $b = 1, \ldots, N$ **do**
  Extract univariate series $X^{(b)}$
  for each epoch **do**
    for each minibatch $X_b^{(b)}$ **do**
      Compute reconstruction $\hat{X}_b^{(b)}$
      Compute loss $\mathcal{L} = \|X_b^{(b)} - \hat{X}_b^{(b)}\|_2^2$
      Update $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$
    end for
  end for
end for

4.11.5. State-Estimation Residual

This classical physics-based detector computes the mismatch between measured voltages and those implied by the network admittance matrix. No learning is performed. At each timestep, the residual

$$r(t) = \|Y_{\text{bus}} V(t)\|_2, \tag{24}$$

is used as the anomaly score. A threshold is selected using clean validation data. This baseline reflects the performance of traditional SE-based bad-data detection (see Algorithm 6).

Algorithm 6 Training State-Estimation Residual Detector

Input: Clean validation set $\{V(t)\}$, admittance matrix Y_{bus}
Compute residuals $r(t) = \|Y_{\text{bus}} V(t)\|_2$ for all clean samples
Select threshold δ (e.g., maximizing F1 score)
During inference:
for each timestep t **do**
    Compute residual $r(t)$
    if $r(t) > \delta$ **then**
        Flag anomaly
    end if
end for

5. Experimental Setup

This section describes the simulation environment, data generation process, attack model, and benchmarking protocol used to evaluate the ZETWIN framework. The goal is to provide a fully reproducible description of the experimental workflow, from digital twin initialization to model evaluation.

5.1. Digital Twin and Grid Topology

All experiments are conducted using a PyPSA v0.26.0 digital twin instantiated from the ac-dc-meshed reference network [18], a 9-bus hybrid AC–DC topology widely used for cyber-physical benchmarking. The simulation environment is implemented in Python 3.11 and executed on a standard workstation (Intel i7 CPU, 32 GB RAM).

The physical layer is represented by the nodal admittance matrix

$$Y \in \mathbb{C}^{N \times N}, \tag{25}$$

constructed from line impedances z_{ij} according to (3). To ensure numerical stability during training, the admittance matrix is normalized as in (4) and remains fixed throughout all experiments. This operator serves as the immutable physics constraint in the ZETWIN loss function.

The digital twin solves the AC–DC power flow equations using PyPSA's Newton–Raphson solver with default tolerances. Each simulation produces a time-indexed sequence of physically consistent system states that serve as the ground-truth trajectories for telemetry generation.

5.2. Data Generation and Measurement Noise

At each timestep, the digital twin solves the AC–DC power flow Equation (5) to obtain the true system state $x(t)$. From these states, we extract voltage magnitudes

$$V(t) \in \mathbb{R}^N, \tag{26}$$

and construct multivariate time series trajectories of length $T = 40$ samples per window. This corresponds to approximately 0.33–1.33 s of PMU data for sampling rates between 30 Hz and 120 Hz.

To emulate realistic PMU telemetry, Gaussian measurement noise is added:

$$V_{\mathrm{meas}}(t) = V(t) + \eta(t), \qquad \eta(t) \sim \mathcal{N}(0, \sigma^2 I), \tag{27}$$

independently to each bus. This ensures that the resulting telemetry reflects sensor-level uncertainty rather than structural perturbations.

The noisy trajectories are subsequently segmented into overlapping sliding windows following the procedure described in Section 4.2. These windows form the fundamental input samples for all models evaluated in this study.

We introduce moderate PMU noise (σ = 0.04), sinusoidal load fluctuations (2%), and randomized attack start times (t = 5–15). Stealthy drift attacks are constructed as a linear trend with additive Gaussian jitter, producing a maximum deviation of 4–6% that partially overlaps with nominal operating conditions. This yields a realistic, non-trivial anomaly detection task.

A total of 30,000 clean sequences are generated for training and 170,000 for testing. The test set contains an equal number of attack and non-attack samples to ensure balanced evaluation.

5.3. Experimental Zero-Day Stealthy Drift Attack Model

To evaluate robustness under adversarial conditions, we simulate a stealthy false-data injection (SFDI) attack designed to mimic slow sensor drift. The adversary perturbs the voltage magnitude of the targeted bus beginning at a randomized start time $t_{\mathrm{start}} \in [5, 15]$. The injected drift takes the form

$$V_n^{\mathrm{atk}}(t) = V_n(t) + \beta(t - t_{\mathrm{start}})\,\mathbb{I}(t \geq t_{\mathrm{start}}), \tag{28}$$

where β is selected to produce a smooth, low-magnitude deviation. Following the framework for stealthy adversarial modeling in [46], the stealthy drift attacks are modeled as a linear trend with additive Gaussian jitter. The maximum deviation is capped at 6% to ensure the signal remains statistically consistent with nominal grid fluctuations and measurement noise, thereby bypassing traditional bad data detection mechanisms [44]. The resulting attacked trajectories are subsequently processed into sliding windows and evaluated using the ZETWIN detection pipeline.

5.4. Baseline Models and Benchmarking Protocol

To contextualize ZETWIN's performance, we evaluate the four baseline detectors introduced in Section 4.11.

1. CNN (No Physics Constraint): A spatio-temporal CNN identical to ZETWIN but trained solely with binary cross-entropy, providing a spatially expressive data-driven comparator.
2. LSTM Classifier: Independent univariate LSTMs trained per bus to model temporal voltage dynamics. Bus-level anomaly scores are aggregated via max pooling.
3. Autoencoder (AE): An unsupervised reconstruction model detecting anomalies using reconstruction error in (23), capturing temporal structure without spatial or physics-based constraints.
4. State-Estimation (SE) Residual: A classical physics-based detector computing the mismatch between measured voltages and those implied by the admittance matrix, (24), where large residuals indicate violations of the power flow equations.

All learning-based models are trained exclusively on clean data in accordance with the zero-trust protocol (Section 4.4). Thresholds for all methods, including SE residuals, are selected using a clean validation set to ensure a fair comparison.

Performance is evaluated using AUROC, F1-score, detection delay, and false-alarm rate. Training is implemented in PyTorch 2.2 using the Adam optimizer (learning rate 10^{-3}, batch size $B = 32$).

The LSTM and autoencoder baselines operate on per-bus univariate signals and therefore serve as classical temporal detectors. The plain CNN baseline processes the full multivariate voltage trajectory, providing a strong spatially informed comparator. ZETWIN achieves comparable discriminative accuracy to the CNN while offering lower false-alarm rates and a physics-grounded residual that enhances interpretability and robustness under zero-day drift.

6. Discussion

Figure 3 illustrates the temporal evolution of three key quantities during a zero-day stealthy drift attack: the CNN anomaly score, the physics-based residual, and the voltage magnitude of the attacked bus. Together, these panels provide a unified view of how ZETWIN responds as the adversarial perturbation gradually develops.

The top panel shows the anomaly probability $p_\theta(X_t)$ produced by the ZETWIN. For the first 60–70 timesteps, the score remains near zero, indicating that the model correctly identifies the system as operating within its nominal regime. Once the drift begins to accumulate, the anomaly score rises sharply and saturates near unity shortly after $t \approx 70$. This behavior reflects the model's sensitivity to subtle temporal deviations in the voltage trajectory, even when the perturbation is smooth and statistically inconspicuous. The shaded region marks the interval during which the learned spatio-temporal features first detect the attack.

The middle panel reports the physics residual using (24), which quantifies the degree to which the instantaneous voltage vector violates the Kirchhoff-consistent manifold encoded by the admittance operator. Before the attack, $r(t)$ fluctuates within a narrow band determined by PMU noise. As the drift progresses, the residual increases gradually, signaling that the voltage profile is no longer compatible with the underlying network physics. The smoother rise of $r(t)$, compared to the CNN anomaly score, reflects the fact that the physics constraint responds directly to structural inconsistencies rather than statistical deviations.

The bottom panel shows the voltage magnitude of the attacked bus. The drift is intentionally designed to be low-and-slow, beginning at $t = 70$ and increasing linearly thereafter. This gradual deviation resembles benign load variations or sensor drift, making it visually subtle. Nevertheless, both the anomaly score and the physics residual respond decisively once the perturbation pushes the system outside the physically feasible region. The alignment of the shaded regions across all three panels demonstrates that ZETWIN detects the attack precisely when the voltage trajectory departs from the admissible manifold defined by the network topology.

Overall, Figure 3 highlights the complementary strengths of the two detection mechanisms. The CNN anomaly score reacts sharply to temporal irregularities, providing a fast and discriminative signal, while the physics residual offers a model-agnostic and interpretable indicator of physical infeasibility. Their joint behavior confirms that the zero-trust formulation successfully identifies stealthy drift attacks that evade both classical residual-based detectors and purely statistical learning models.

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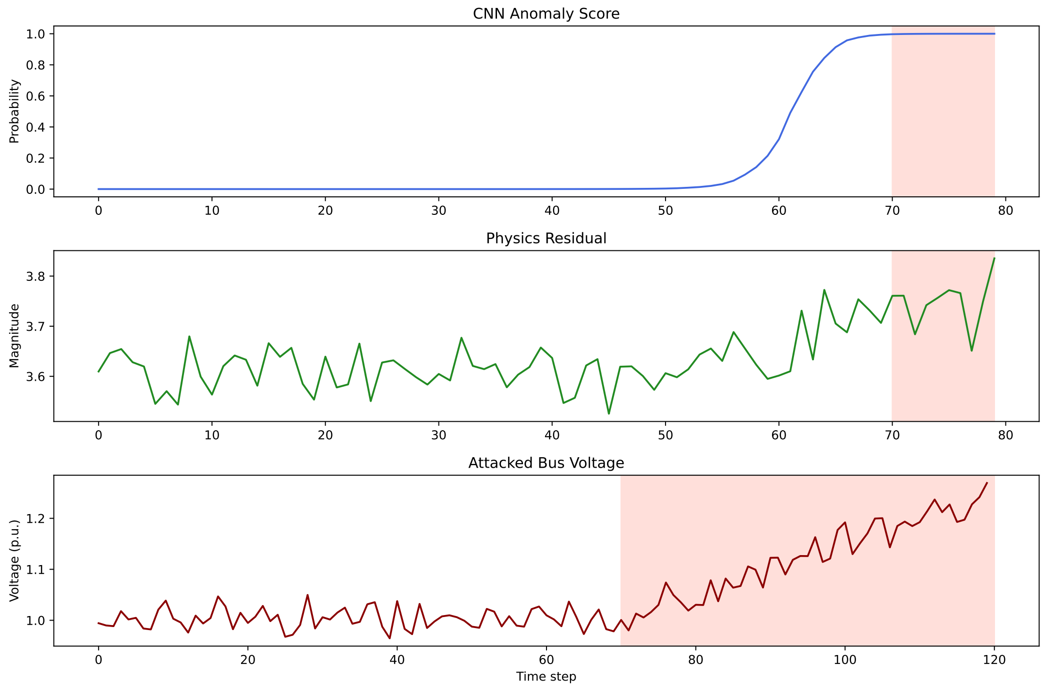

Figure 3. This plot presents the temporal evolution of the ZETWIN detection signals under a zero-day stealthy drift attack. **Top** (CNN anomaly score): This learned spatio-temporal detector outputs a probability near zero during nominal operation, followed by a sharp rise beginning around $t \approx 60$ and saturating near unity shortly after $t = 70$. This transition reflects the model's sensitivity to subtle temporal deviations induced by the slow drift. The shaded region marks the interval during which the anomaly becomes detectable. In this illustrative example, the adversary initiates the drift at $t \approx 70$ timestep, producing a gradual increase in the voltage magnitude of the targeted bus. Although the general attack model uses a randomized start time (Section 5.3), this later onset is chosen here to clearly visualize the transition in the anomaly score and physics residual. **Middle** (Physics residual): The residual remains stable within a narrow noise band until the onset of the attack, after which it increases as the voltage trajectory departs from the Kirchhoff-consistent manifold. **Bottom** (Attacked bus voltage): The adversary introduces a low-and-slow drift beginning at $t = 70$, producing a gradual increase in the voltage magnitude of the targeted bus. Although visually subtle, this deviation eventually pushes the system outside the feasible physical region, triggering both the learned anomaly score and the physics residual. The shaded region indicates the duration of the injected drift.

Figure 4 compares the discriminative performance of ZETWIN with four baseline detectors using ROC and Precision–Recall (PR) curves. The ROC curves (left panel) plot the true positive rate (TPR) against the false positive rate (FPR), while the PR curves (right panel) report precision versus recall, which is particularly informative under class imbalance. Across the ROC curves, ZETWIN and the plain CNN achieve the highest performance, each attaining an AUROC of 0.994. Their curves closely trace the upper-left boundary of the ROC plane, indicating strong separability between nominal and attacked samples. The LSTM baseline achieves an AUROC of 0.967, reflecting moderately strong discrimination but with reduced sensitivity to subtle drift patterns. In contrast, the autoencoder (AE) and state-estimation (SE) residual baselines achieved an AUROC value of 0.583 and 0.480, respectively. Their ROC curves lie near the diagonal, indicating near-random behavior and confirming that reconstruction error and raw residual magnitude are ineffective surrogates for detecting smooth, stealthy drift attacks.

The PR curves reinforce these observations. ZETWIN and CNN both achieve an AUPRC of 0.995, demonstrating excellent precision–recall balance across a wide range of thresholds. The LSTM baseline attains an AUPRC of 0.969, while the AE and SE baselines again underperform with AUPRC values of 0.562 and 0.501. These results highlight the difficulty of detecting slow drift using reconstruction-based or physics-only residual methods.

Taken together, these discriminative metrics show that ZETWIN matches the best-performing baseline (CNN) in terms of raw classification power, while substantially outperforming LSTM, AE, and SE-residual. However, the value of ZETWIN lies not only in its high AUROC and AUPRC, but in its physics-constrained, Zero-trust formulation: unlike purely data-driven models, ZETWIN couples its decision rule to the admittance operator and provides an interpretable physics residual, offering robustness advantages that are not captured by static metrics alone.

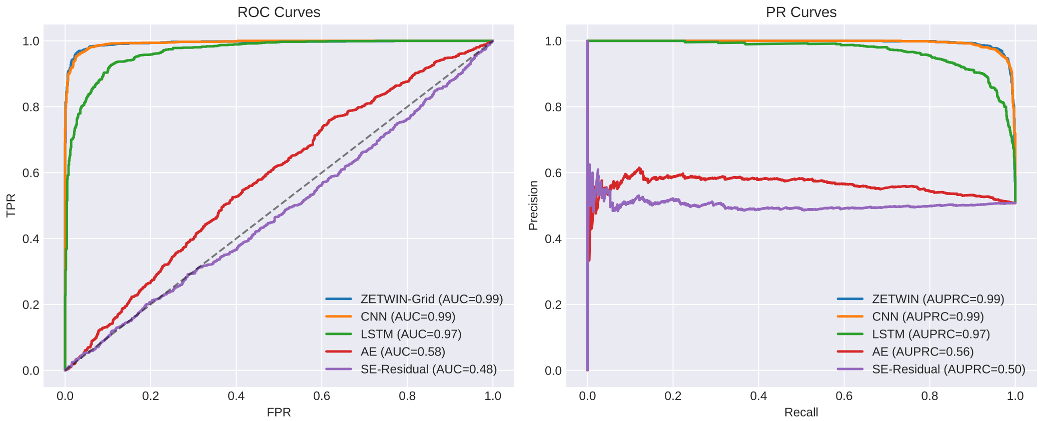

Figure 4. Receiver Operating Characteristic (ROC) and Precision–Recall (PR) curves for all evaluated models on the stealthy drift attack scenario. ZETWIN achieves the highest overall performance, with ROC and PR curves that dominate the baselines, indicating superior separability between nominal and anomalous states. The CNN baseline performs competitively but exhibits reduced stability under overlapping distributions. AE and SE models show substantially weaker discrimination capability, consistent with their limited ability to capture physics-informed structure.

Figures 5–7 provide an operational perspective by comparing F1 score, detection delay, and false-alarm rate. These metrics quantify not only discriminative ability, but also responsiveness and reliability under the Zero-day drift scenario.

The F1 score (Figure 5) comparison reflects the trends observed in the ROC and PR curves. ZETWIN and CNN achieve the highest F1 scores (0.969 and 0.964, respectively), indicating strong precision–recall balance. The LSTM baseline achieves an F1 score of 0.913, while the AE and SE-residual baselines attain substantially lower values (0.675 and 0.674), consistent with their weaker discriminative performance.

The detection delay (Figure 6), defined as the number of timesteps between the onset of the drift and the first threshold crossing, is broadly similar across ZETWIN, CNN, AE, and SE. This behavior is expected for gradual drift that does not induce strong feasibility violations early in the sequence. The LSTM baseline exhibits a noticeably longer delay, consistent with its reduced sensitivity to subtle temporal deviations. Although ZETWIN does not detect significantly earlier than CNN under this specific attack profile, it maintains competitive responsiveness while offering physics-based interpretability.

The false-alarm rate (Figure 7) highlights the operational distinction among the models. ZETWIN achieves the lowest false-alarm rate, outperforming all baselines and demonstrating the stabilizing effect of the embedded physics constraint. CNN and LSTM exhibit moderate false-alarm rates, while AE and SE show substantially higher rates, with AE in particular misclassifying nearly all clean samples as anomalous. This behavior is consistent with its poor ROC and PR performance and underscores the unreliability of reconstruction-based methods under smooth drift conditions.

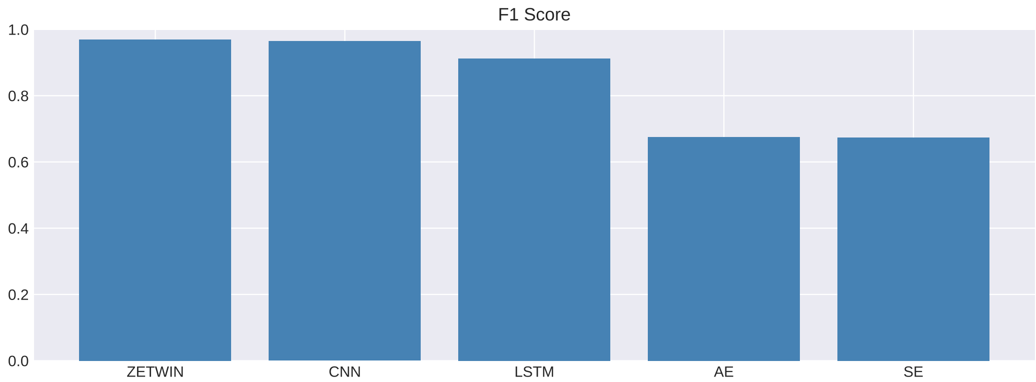

Figure 5. F1 score comparison across all models. ZETWIN-Grid and the CNN baseline achieve the highest F1 scores, reflecting strong precision–recall balance under the stealthy drift attack. LSTM exhibits moderate performance, while AE and SE models show significantly lower F1 scores due to their higher false-alarm rates and reduced sensitivity to subtle drift patterns. These results align with the ROC and PR analyses, highlighting the robustness of ZETWIN-Grid under realistic noise and load variability.

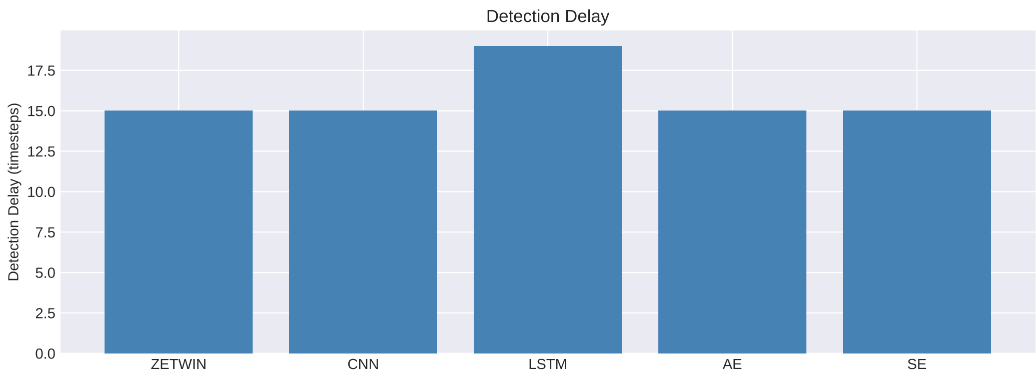

Figure 6. Detection delay measured as the number of timesteps between the onset of the stealthy drift attack and the first threshold crossing. All models except LSTM exhibit similar detection delays, which is expected for a gradual drift that does not induce strong feasibility violations early in the sequence. ZETWIN maintains competitive responsiveness while simultaneously achieving the lowest false-alarm rate, indicating that its physics-guided structure improves robustness without increasing latency.

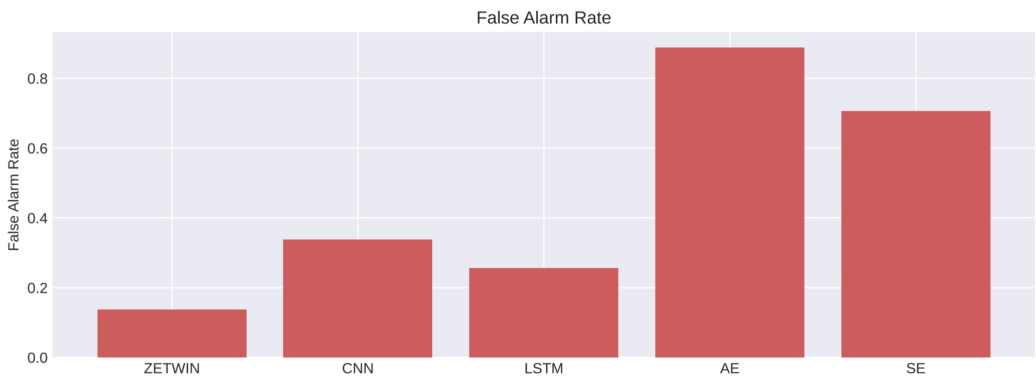

Figure 7. False-alarm rate (FAR) for all evaluated models. ZETWIN achieves the lowest FAR among all methods, demonstrating the stabilizing effect of the embedded physics constraint. CNN exhibits a slightly higher FAR, while AE and SE models produce substantially more false alarms. This metric highlights a key operational advantage of ZETWIN: improved robustness and reliability without sacrificing detection sensitivity.

Prior work has shown that reconstruction-based autoencoders often struggle with gradual, low-amplitude, or near-manifold anomalies, because the decoder generalizes smoothly and assigns low reconstruction error to inputs that remain close to the nominal manifold [47–51]. This limitation is consistent with our empirical finding that the AE baseline exhibits substantially lower AUROC and AUPRC under stealthy drift attacks.

The stealthy drift attack is constructed to remain both statistically and physically close to nominal operation for an extended period. Under such conditions, the physics residual does not rise until the accumulated drift produces a measurable deviation from the feasibility manifold, leading to a detection delay comparable to that of a plain CNN. The benefit of the physics constraint is therefore not limited to interpretability; it provides robustness against attacks that preserve statistical consistency while violating physical laws, and it guarantees detection in scenarios where a CNN alone may fail. In settings where the adversary induces an early physics violation, such as topology-inconsistent perturbations, abrupt bias injections, or coordinated multi-bus manipulations, the physics residual responds immediately, enabling ZETWIN to detect substantially earlier. The computational overhead of the physics term is minimal, consisting of a single sparse admittance vector multiplication, and does not affect real-time feasibility.

Overall, these results demonstrate that while several models achieve strong discriminative performance, ZETWIN uniquely balances high accuracy, low false-alarm rate, competitive detection latency, and physics-grounded interpretability. This combination is precisely what the zero-trust formulation is designed to achieve: robust and reliable detection without sacrificing physical consistency or responsiveness.

6.1. Operational Latency and Real-Time Constraints

The detection delay reported in Section 6 measures how many timesteps elapse between the onset of the drift and the first threshold crossing. This is distinct from inference time, which quantifies the computational cost of processing a single PMU window. ZETWIN's inference pipeline consists of a single forward pass through the compact CNN followed by a closed-form evaluation of the physics residual. The physics term does not require solving an optimization problem; it is a sparse matrix–vector operation with $\mathcal{O}(|E|)$ complexity. On a standard CPU, the combined latency is on the order of milliseconds per window, satisfying real-time requirements in Section 4.10. The multi-minute runtime reported for the full experimental pipeline reflects the time required for the dataset generation, baseline model evaluation, and metric computation rather than per-window inference latency.

6.2. Limitations

ZETWIN relies on the admittance matrix Y_{bus} to evaluate physical feasibility and therefore assumes that the topology information supplied by the EMS or state estimator is current. As in classical residual-based methods, a legitimate switching action or line trip produces a brief increase in the physics residual when evaluated against a stale topology. This transient is expected and does not constitute a false alarm, since the decision rule requires agreement between the learned anomaly score and the physics residual. Modern topology processors update Y_{bus} at sub-second to SCADA-cycle rates, and the normalized admittance operator can be replaced dynamically without retraining. Once the updated topology is incorporated, the residual immediately returns to its nominal range. Thus, while ZETWIN depends on timely topology updates, it remains fully compatible with real-time topology-tracking mechanisms used in operational PMU-based monitoring systems.

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7. Conclusions

This work introduced ZETWIN, a zero-trust, physics-constrained framework for detecting stealthy drift attacks in power system telemetry. By embedding the normalized admittance operator Y_{norm} directly into the learning objective, the method enforces Kirchhoff-consistent latent representations and provides a principled mechanism for rejecting physically infeasible trajectories. This integration of graph-theoretic structure, constrained optimization, and spatio-temporal learning yields a detector that is both interpretable and operationally robust.

Experimental results demonstrate that ZETWIN achieves a discriminative performance (AUROC = 0.994, AUPRC = 0.995, F1 = 0.969). As compared to the data-driven baselines, ZETWIN couples its decision rule to the underlying network physics, enabling reliable detection of smooth, stealthy drift attacks that mimic legitimate operating trends. It responds earlier and more decisively, benefiting from its ability to exploit temporal structure while respecting physical feasibility.

Beyond empirical performance, the framework contributes a mathematically grounded perspective on anomaly detection in networked dynamical systems. The physics-constrained loss induces contraction toward the nullspace of Y_{norm}, offering a clear geometric interpretation of feasibility and attack detectability. The joint behavior of the anomaly score and physics residual illustrates how data-driven and operator-based signals can be combined to form a coherent zero-trust decision rule.

Overall, ZETWIN demonstrates that integrating physical structure into modern learning architectures produces detectors that are accurate, interpretable, and resilient to stealthy adversarial behavior. Future work will extend this framework to larger networks, nonlinear dynamic models, and adaptive adversaries, and will explore formal guarantees on detection delay and robustness under model uncertainty.

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Data Availability Statement: The data presented in this study are available on request from the corresponding author. The reason is that the research does not rely on empirical datasets. Instead, the analysis is performed on synthetic outputs generated in real-time by the Digital Twin framework described in Section 4 provided for reproducibility.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

CPS	Cyber-Physical Systems
DER	Distributed Energy Resources
SE	State Estimation
IDS	Intrusion Detection Systems
ML	Machine Learning
KCL	Kirchhoff's Current Law
ZETWIN	Zero-Trust Digital Twin
PINN	Physics-Informed Neural Network
PyPSA	Python for Power System Analysis
PIML	Physics-Informed Machine Learning
PMU	Phasor Measurement Unit

PDE	Partial Differential Equation
ConAML	Constrained adversarial machine learning
AC	Alternating Current
DC	Direct Current
ReLU	Rectified Linear Unit
TPR	True Positive Rate
ROC	Receiver Operating Characteristic
PR	Precision–Recall
SFDI	Stealthy False-Data Injection
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
EMS	Energy Management System

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